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Integrated Services Digital Network (ISDN); Three-Party (3PTY) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 5: Test Suite Structure and Test Purposes (TSS&TP) specification for the network

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Foreword

This European Telecommunication Standard (ETS) has been produced by the Signalling Protocols and Switching (SPS) Technical Committee of the European Telecommunications Standards Institute (ETSI).

This ETS is part 5 of a multi-part standard covering the Digital Subscriber Signalling System No. one (DSS1) protocol specification for the Integrated Services Digital Network (ISDN) Three-Party (3PTY) supplementary service, as described below:

Part 1: "Protocol specification";

Part 2: "Protocol Implementation Conformance Statement (PICS) proforma specification";

Part 3: "Test Suite Structure and Test Purposes (TSS&TP) specification for the user";

Part 4: "Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma specification for the user";

Part 5: "TSS&TP specification for the network";

Part 6: "ATS and partial PIXIT proforma specification for the network".

Transposition dates	
Date of adoption of this ETS:	30 August 1996
Date of latest announcement of this ETS (doa):	31 December 1996
Date of latest publication of new National Standard or endorsement of this ETS (dop/e):	30 June 1997
Date of withdrawal of any conflicting National Standard (dow):	30 June 1997

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1 Scope

This fifth part of ETS 300 188 specifies the Test Suite Structure and Test Purposes (TSS&TP) for the Network side of the T reference point or coincident S and T reference point (as defined in ITU-T Recommendation I.411 [7]) of implementations conforming to the stage three standard for the Three-Party (3PTY) supplementary service for the pan-European Integrated Services Digital Network (ISDN) by means of the Digital Subscriber Signalling System No. one (DSS1) protocol, ETS 300 188-1 [1].

A further part of this ETS specifies the Abstract Test Suite (ATS) and partial Protocol Implementation eXtra Information for Testing (PIXIT) proforma based on this ETS. Other parts specify the TSS&TP and the ATS and partial PIXIT proforma for the User side of the T reference point or coincident S and T reference point of implementations conforming to ETS 300 188-1 [1].

2 Normative references

This ETS incorporates by dated and undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

[1] ETS 300 188-1 (1993): "Integrated Services Digital Network (ISDN); Three-Party (3PTY) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[2] ETS 300 188-2 (1995): "Integrated Services Digital Network (ISDN); Three-Party (3PTY) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 2: Protocol Implementation Conformance Statement (PICS) proforma specification".

[3] ISO/IEC 9646-1: "Information Technology - OSI Conformance Testing Methodology and Framework; Part 1: General Concepts".

[4] ISO/IEC 9646-2: "Information Technology - OSI Conformance Testing Methodology and Framework; Part 2: Abstract Test Suite specification".

[5] ISO/IEC 9646-3: "Information Technology - OSI Conformance Testing Methodology and Framework; Part 3: The Tree and Tabular Combined Notation".

[6] ETS 300 196-1 (1993): "Integrated Services Digital Network (ISDN); Generic functional protocol for the support of supplementary services; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[7] ITU-T Recommendation I.411 (1993): "ISDN user-network interfaces - Reference configurations".

[8] ETS 300 102-1: "Integrated Services Digital Network (ISDN); User-network interface layer 3; Specifications for basic call control".

[9] ITU-T Recommendation I.112 (1993): "Vocabulary and terms for ISDNs".

[10] CCITT Recommendation E.164 (1991): "Numbering plan for the ISDN era".

[11] ITU-T Recommendation I.210 (1993): "Principles of the telecommunication services supported by an ISDN and the means to describe them".

3 Definitions

For the purposes of this ETS, the following definitions apply:

3.1 Definitions related to conformance testing

abstract test case: Refer to ISO/IEC 9646-1 [3].

Abstract Test Suite (ATS): Refer to ISO/IEC 9646-1 [3].

active test: A test case where the IUT is required to send a particular message, but not in reaction to a received message. This would usually involve the use of PIXIT information to see how this message can be generated and quite often is specified in an ATS using an implicit send event.

Implementation Under Test (IUT): Refer to ISO/IEC 9646-1 [3].

implicit send event: Refer to ISO/IEC 9646-3 [5].

lower tester: Refer to ISO/IEC 9646-1 [3].

passive test: A test case where the IUT is required to respond to a protocol event (e.g. received message) with another protocol event (sends message) and normally does not require any special operator intervention such as is associated with the implicit send event.

point of control and observation: Refer to ISO/IEC 9646-1 [3].

Protocol Implementation Conformance Statement (PICS): Refer to ISO/IEC 9646-1 [3].

PICS proforma: Refer to ISO/IEC 9646-1 [3].

Protocol Implementation eXtra Information for Testing (PIXIT): Refer to ISO/IEC 9646-1 [3].

PIXIT proforma: Refer to ISO/IEC 9646-1 [3].

system under test: Refer to ISO/IEC 9646-1 [3].

Test Purpose (TP): Refer to ISO/IEC 9646-1 [3].

3.2 Definitions related to ETS 300 188-1

3PTY Active state: An occurrence of the 3PTY supplementary service exists.

3PTY Await Hold state: A Hold function should be initiated by the user prior to ending of the 3PTY supplementary service.

3PTY Await Hold And Retrieve state: A Hold function and a Retrieve function should be initiated by the user prior to ending of the 3PTY supplementary service.

3PTY Await Retrieve state: A Retrieve function should be initiated by the user prior to ending of the 3PTY supplementary service.

3PTY Idle state: An occurrence of the 3PTY supplementary service has not been requested.

Call Held auxiliary state: See ETS 300 196-1 [6], subclause 7.1.2.

call reference: See ETS 300 102-1 [8], subclause 4.3.

component: See ETS 300 196-1 [6], subclause 11.2.2.1.

Idle auxiliary state: See ETS 300 196-1 [6], subclause 7.1.2.

Integrated Services Digital Network (ISDN): See ITU-T Recommendation I.112 [9], definition 308.

invoke component: See ETS 300 196-1 [6], subclause 11.2.2.1.

ISDN number: A number conforming to the numbering and structure specified in CCITT Recommendation E.164 [10].

network: The DSS1 protocol entity at the Network side of the user-network interface where a T reference point or coincident S and T reference point applies.

network (S/T): The DSS1 protocol entity at the network side of the user-network interface where a coincident S and T reference point applies.

network (T): The DSS1 protocol entity at the Network side of the user-network interface where a T reference point applies (Network connected to Private ISDN).

return error component: See ETS 300 196-1 [6], subclause 11.2.2.1.

return result component: See ETS 300 196-1 [6], subclause 11.2.2.1.

served user: The served user is the user who invokes the 3PTY supplementary service.

service; telecommunication service: See ITU-T Recommendation I.112 [9], definition 201.

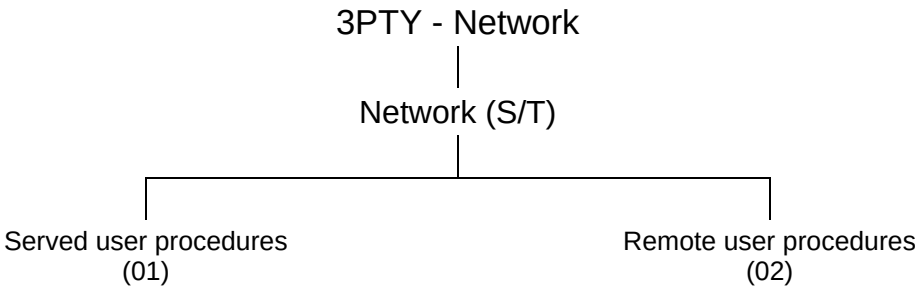
supplementary service: See ITU-T Recommendation I.210 [11], subclause 2.4.

4 Abbreviations

For the purposes of this ETS, the following abbreviations apply:

(Held)	Call Held auxiliary state
(Idle)	Idle auxiliary state
3PTY	Three-Party
ATM	Abstract Test Method
ATS	Abstract Test Suite
CEI	Connection Endpoint Identifier
CR1	Call Reference for a call in the Active call state and Idle auxiliary state
CR2	Call Reference for a call in the Active call state and Call Held auxiliary state
DSS1	Digital Subscriber Signalling System No. one
ISDN	Integrated Services Digital Network
IUT	Implementation Under Test
N00	Idle call state
N04	Call Delivered call state
N10	Active call state
N12	Disconnect Indication call state
N19	Release Request call state
TP	Test Purpose
TSS	Test Suite Structure

5 Test Suite Structure (TSS)



NOTE: Numbers in brackets represent group numbers and are used in TP identifiers.

Figure 1: Test suite structure

6 Test Purposes (TP)

6.1 Introduction

For each test requirement a TP is defined.

6.1.1 TP naming convention

TPs are numbered, starting at 001, within each group. Groups are organized according to the TSS. Additional references are added to identify the actual test suite and whether it applies to the network or the user (see table 1).

Table 1: TP identifier naming convention scheme

Identifier:	**<ss>_<iut><group>_<nnn>**			
<ss>	=	supplementary service:	e.g. "3PTY"	
<iut>	=	type of IUT:	U	User
			N	Network
<group>	=	group	2 digit field representing group reference according to TSS	
<nnn>	=	sequential number	(001-999)	

6.1.2 Source of TP definition

The test purposes are based on ETS 300 188-1 [1], clauses 9, 10 and 14.

6.1.3 TP structure

Each TP has been written in a manner which is consistent with all other TPs. The intention of this is to make the TPs more readable and checkable. A particular structure has been used and this is illustrated in table 2. This table should be read in conjunction with any TP, i.e. use a TP as an example to fully understand the table.

Table 2: Structure of a single TP

TP part	Text	Example
Header	**<Identifier>** *tab* **<paragraph number in base ETS>** *tab* **<type of test>** *tab* **<condition>** *CR*	see table 1 **subclause 0.0.0** **valid, invalid, inopportune** **mandatory, optional, conditional**
Stimulus	Ensure that the IUT in the <supplementary service state> and with CR1 in <basic call state> (<auxiliary state>) and with CR2 in <basic call state> (<auxiliary state>) <trigger> *see below for message structure* *or* <goal>	 3PTY Idle state and N10(idle), N10(held), etc. -"- receiving a XXXX message to request a ....
Reaction	<action> <conditions> *if the action is sending* *see below for message structure* <next action>, *etc.* and enters <supplementary service state> *and/or* and remains in the same state(s) *or* and enters state <state> with CR<number(s)>	sends, saves, does, etc. using en bloc sending, ...
Message structure	<message type> message containing a *a)* <info element> information element with *b)* a <field name> encoded as *or* including <coding of the field> and *back to a or b,*	SETUP, FACILITY, CONNECT, … Bearer capability, Facility, ...
NOTE:	Text in italics will not appear in TPs and text between <> is filled in for each TP and may differ from one TP to the next.	

6.1.4 Test strategy

As the base standard ETS 300 188-1 [1] contains no explicit requirements for testing, the TPs were generated as a result of an analysis of the base standard and the PICS specification ETS 300 188-2 [2]. The criteria applied include the following:

- only the requirements from the point of view of the T or coincident S and T reference point are considered;

- whether or not a test case can be built from the TP is not considered.

6.2 Network TPs for 3PTY

NOTE: All calls have the same Connection Endpoint Identifier (CEI) value.

6.2.1 Served user procedures

3PTY_N01_001 subclause 9.2.1.1, 2nd paragraph valid mandatory
Ensure that the IUT in the 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with a Begin3PTY invoke component,
responds with a FACILITY message containing a Facility information element with a Begin3PTY return result component for CR2.

3PTY_N01_002 subclause 9.2.1.2, 1st & 2nd paragraph inopportune mandatory
Ensure that the IUT in the 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with a Begin3PTY invoke component and the supplementary service has not been subscribed,
responds with a FACILITY message containing a Facility information element with a Begin3PTY return error component coded as "notSubscribed" for CR2 and remains in the same state.

3PTY_N01_003 subclause 9.2.1.2, 1st & 2nd paragraph inopportune mandatory
Ensure that the IUT, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR1 containing a Facility information element with a Begin3PTY invoke component,

responds with a FACILITY or INFORMATION message containing a Facility information element with a Begin3PTY return error component coded as "invalidCallState" and remains in the same state for CR1.

3PTY_N01_004 subclause 9.2.1.2, 1st & 2nd paragraph inopportune mandatory
Ensure that the IUT, with CR1 in call state N04 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR1 containing a Facility information element with a Begin3PTY invoke component,

responds with a FACILITY, PROGRESS, CONNECT or INFORMATION message containing a Facility information element with a Begin3PTY return error component coded as "invalidCallState" and continues normal call handling for CR1.

NOTE 1: This is the case where "no Active-Idle connection exists for CEI value".

3PTY_N01_005 subclause 9.2.1.2, 1st & 2nd paragraph inopportune optional
Ensure that the IUT, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held) and a third in call state N10 (Idle), on receipt of a FACILITY message for CR1 containing a Facility information element with a Begin3PTY invoke component,

responds with a FACILITY or INFORMATION message containing a Facility information element with a Begin3PTY return error component coded as "invalidCallState" and remains in the same state for CR1.

Selection: IUT must be able to support 3 calls.

NOTE 2: This is the case where "more than one Active-Idle connection exists for CEI value".

3PTY_N01_006 subclause 9.2.1.2, 1st & 2nd paragraph inopportune mandatory
Ensure that the IUT in 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with a Begin3PTY invoke component, and a three-way bridge is not available for use,

responds with a FACILITY or INFORMATION message containing a Facility information element with a Begin3PTY return error component coded as "resourceUnavailable" and remains in the same states.

3PTY_N01_007 subclause 9.2.1.2, 1st & 2nd paragraph inopportune mandatory
Ensure that the IUT in 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with a Begin3PTY invoke component and a three-way bridge already exists for one or both of the connections (relating to the same served user),

responds with a FACILITY or INFORMATION message containing a Facility information element with a Begin3PTY return error component coded as "supplementaryServiceInteractionNotAllowed" and remains in the same state for CR2.

3PTY_N01_008 subclause 9.2.1.2, 1st & 2nd paragraph inopportune mandatory
Ensure that the IUT in 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with a Begin3PTY invoke component and the provision of the service is precluded by a procedure within ETS 300 195,

responds with a FACILITY or INFORMATION message containing a Facility information element with a Begin3PTY return error component coded as "supplementaryServiceInteractionNotAllowed" and remains in the same state for CR2.

3PTY_N01_009 subclause 9.2.1.2, 4th paragraph invalid mandatory
Ensure that the IUT in 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a CONNECT ACKNOWLEDGE message for CR2 containing a Facility information element with a Begin3PTY invoke component,

responds with a FACILITY or INFORMATION message containing a Facility information element with a Begin3PTY return error component coded as "invalidCallState" and remains in the same state for CR2.

3PTY_N01_010 subclause 9.2.1.2, 4th paragraph invalid mandatory
Ensure that the IUT in 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a HOLD message for CR1 containing a Facility information element with a Begin3PTY invoke component,

responds with a HOLD ACKNOWLEDGE message containing a Facility information element with a Begin3PTY return error component coded as "invalidCallState" and enters the Call Held auxiliary state for CR1.

3PTY_N01_011 subclause 9.2.1.2, 6th paragraph inopportune mandatory
Ensure that the IUT in 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with a Begin3PTY reject component,

takes no action and remains in the same state.

3PTY_N01_012 subclause 9.2.1.2, 7th paragraph inopportune mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a RETRIEVE message for CR2,

responds with a RETRIEVE REJECT message with cause #29 "Facility rejected" and remains in the same state.

3PTY_N01_013 subclause 9.2.2.1 a) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a DISCONNECT message for CR1,

sends a RELEASE message and enters N19 for CR1 and enters the 3PTY Await Retrieve state.

3PTY_N01_014 subclause 9.2.2.1 a) valid mandatory
Ensure that the IUT in 3PTY Await Retrieve state, with CR1 in call state N00 and CR2 in call state N10 (Held), after clearing was initiated by the receipt of a DISCONNECT message for CR1, on receipt of a RETRIEVE message for CR2,

sends a RETRIEVE ACKNOWLEDGE message and enters N10(Idle) for CR2 and enters the 3PTY Idle state.

3PTY_N01_015 subclause 9.2.2.1 b) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a DISCONNECT message for CR2,

sends a RELEASE message and enters N19 for CR2 and enters the 3PTY Idle state.

3PTY_N01_016 subclause 9.2.3.1, 2nd paragraph valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of DISCONNECT messages for both calls but the first one is for CR1,

sends a RELEASE message and enters state N19 for CR1.

NOTE 3: First DISCONNECT message is for the Active-Idle call.

3PTY_N01_017 subclause 9.2.3.1, 2nd paragraph valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of DISCONNECT messages for both calls but the first one is for CR2,

sends a RELEASE message and enters state N19 for CR2.

NOTE 4: The first DISCONNECT message is for the Active-Held call.

3PTY_N01_018 subclause 9.2.4.1 a) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an End3PTY invoke component,

responds with a FACILITY message for CR2 containing a Facility information element with an End3PTY return result component and enters the 3PTY Await Hold And Retrieve state.

NOTE 5: Private communication with the Active-Held remote user.

3PTY_N01_019 subclause 9.2.4.1 a) valid mandatory
Ensure that the IUT in the 3PTY Await Hold And Retrieve state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a HOLD message for CR1,
responds with a HOLD ACKNOWLEDGE message for CR1 and enters N10 (Held) state for CR1 and enters the 3PTY Await Retrieve state.

NOTE 6: Private communication with the Active-Held remote user.

3PTY_N01_020 subclause 9.2.4.1 a) valid mandatory
Ensure that the IUT in 3PTY Await Retrieve state, with CR1 in call state N10 (Held) and CR2 in call state N10 (Held), on receipt of a RETRIEVE message for CR2,
responds with a RETRIEVE ACKNOWLEDGE message for CR2 and enters N10 (Idle) state for CR2 and enters the 3PTY Idle state.

NOTE 7: Private communication with the Active-Held remote user.

3PTY_N01_021 subclause 9.2.4.1 a) & Fig 2.5 & 2.6 valid mandatory
Ensure that the IUT in the 3PTY Await Retrieve state, with CR1 in call state N00, after clearing of CR1 while in the 3PTY Await Hold And Retrieve state, and CR2 in call state N10 (Held), on receipt of a DISCONNECT message for CR2,
sends a RELEASE message and enters N19 for CR2 and enters the 3PTY Idle state.

NOTE 8: Private communication with the Active-Held remote user.

3PTY_N01_022 subclause 9.2.4.1 a) & Fig 2.5 & 2.6 valid mandatory
Ensure that the IUT in the 3PTY Await Retrieve state, with CR1 in call state N10 (Held) and CR2 in call state N10 (Held), on receipt of a DISCONNECT message for CR2,
sends a RELEASE message and enters N19 for CR2 and enters the 3PTY Idle state.

NOTE 9: The 3PTY Await Retrieve state may be entered when establishing a private communication with a remote user whose call is in the Held auxiliary state. Normally this procedure involves the holding of the Idle call and retrieving of the Held call. The 3PTY Await Retrieve state is entered once the Idle call is put on hold and before the Held call is retrieved.

3PTY_N01_023 subclause 9.2.4.1 a) & Fig 2.6 valid mandatory
Ensure that the IUT in the 3PTY Await Hold And Retrieve state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a DISCONNECT message for CR1,
sends a RELEASE message and enters N19 for CR1 and enters the 3PTY Await Retrieve state.

NOTE 10: The 3PTY Await Hold And Retrieve state is only entered when there is an attempt to establish a private communication with a remote user whose call is in the Held auxiliary state.

3PTY_N01_024 subclause 9.2.4.1 a) & Fig 2.6 valid mandatory
Ensure that the IUT in the 3PTY Await Hold And Retrieve state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a DISCONNECT message for CR2,
sends a RELEASE message and enters N19 for CR2 and enters the 3PTY Idle state.

NOTE 11: The 3PTY Await Hold And Retrieve state is only entered when there is an attempt to establish a private communication with a remote user whose call is in the Held auxiliary state.

3PTY_N01_025 subclause 9.2.4.1 a) & Fig 2.7 valid mandatory
Ensure that the IUT in the 3PTY Await Hold state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Idle), the call associated with CR2 having been retrieved from the call Held state, on receipt of a DISCONNECT message for CR1,

sends a RELEASE message and enters N19 for CR1, and enters the 3PTY Idle state.

NOTE 12: The 3PTY Await Hold state is only entered when there is an attempt to establish a private communication with a remote user whose call is in the Held auxiliary state. Normally this procedure involves the holding of the Idle call and retrieving of the Held call. The 3PTY Await Hold state is entered once the Held call is retrieved and before the Idle call is put on hold.

3PTY_N01_026 subclause 9.2.4.1 b) valid mandatory
Ensure that the IUT in 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR1 containing a Facility information element with an End3PTY invoke component,

responds with a FACILITY message for CR1 containing a Facility information element with an End3PTY return result component and enters the 3PTY Idle state.

NOTE 13: Private communication with the Active-Idle remote user.

3PTY_N01_027 subclause 9.2.4.2, 1st paragraph inopportune mandatory
Ensure that the IUT in 3PTY Idle state (i.e. no three-way conversation exists), with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an End3PTY invoke component,

responds with a FACILITY or INFORMATION message for CR2 containing a Facility information element with an End3PTY return error component coded as "invalidCallState" and remains in the same state.

3PTY_N01_028 subclause 9.2.4.2, 1st paragraph inopportune mandatory
Ensure that the IUT in 3PTY Await Hold And Retrieve state (i.e. no three-way conversation exists), with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an End3PTY invoke component,

responds with a FACILITY or INFORMATION message for CR2 containing a Facility information element with an End3PTY return error component coded as "invalidCallState" and remains in the same state.

3PTY_N01_029 subclause 9.2.4.2, 1st paragraph inopportune mandatory
Ensure that the IUT in 3PTY Await Retrieve state (i.e. no three-way conversation exists), with CR1 in call state N10 (Held) or N00 and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR2 containing a Facility information element with an End3PTY invoke component,

responds with a FACILITY or INFORMATION message for CR2 containing a Facility information element with an End3PTY return error component coded as "invalidCallState" and remains in the same state.

3PTY_N01_030 subclause 9.2.4.2, 1st paragraph inopportune mandatory
Ensure that the IUT in 3PTY Await Hold state (i.e. no three-way conversation exists), with CR1 in call state N10 (Idle) and CR2 in call state N10 (Idle), on receipt of a FACILITY message for CR2 containing a Facility information element with an End3PTY invoke component,

responds with a FACILITY or INFORMATION message for CR2 containing a Facility information element with an End3PTY return error component coded as "invalidCallState" and remains in the same state.

3PTY_N01_031 subclause 9.2.4.2, 1st paragraph inopportune mandatory
Ensure that the IUT in the 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of a FACILITY message for CR1 containing a Facility information element with an End3PTY invoke component but no three-way conversation exists,

responds with a FACILITY or INFORMATION message for CR1 containing a Facility information element with an End3PTY return error component coded as "invalidCallState" and remains in the same state.

3PTY_N01_032 subclause 9.2.4.2, 2nd paragraph inopportune mandatory
Ensure that the IUT in the 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of an INFORMATION message for CR2 containing a Facility information element with an End3PTY invoke component but no three-way conversation exists,

responds with a FACILITY or INFORMATION message for CR2 containing a Facility information element with an End3PTY return error component coded as "invalidCallState" and remains in the same state.

3PTY_N01_033 subclause 9.2.4.2, 2nd paragraph inopportune mandatory
Ensure that the IUT in the 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of an INFORMATION message for CR1 containing a Facility information element with an End3PTY invoke component but no three-way conversation exists,

responds with a FACILITY or INFORMATION message for CR1 containing a Facility information element with an End3PTY return error component coded as "invalidCallState" and remains in the same state.

3PTY_N01_034 subclause 9.2.4.2, 3rd paragraph inopportune mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of an INFORMATION message for CR1 containing a Facility information element with an End3PTY invoke component and a three-way conversation exists,

responds with a FACILITY or INFORMATION message for CR1 containing a Facility information element with an End3PTY return error component coded as "invalidCallState" and remains in the same state.

3PTY_N01_035 subclause 9.2.4.2, 3rd paragraph inopportune mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), on receipt of an INFORMATION message for CR2 containing a Facility information element with an End3PTY invoke component and a three-way conversation exists,

responds with a FACILITY or INFORMATION message for CR2 containing a Facility information element with an End3PTY return error component coded as "invalidCallState" and remains in the same state.

3PTY_N01_036 subclause 9.2.4.2, 7th paragraph valid mandatory
Ensure that the IUT in the 3PTY Await Hold And Retrieve state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), after sending a FACILITY message for CR2 containing a Facility information element with an End3PTY return result component, on receipt of a FACILITY message for CR2 containing a Facility information element with an End3PTY reject component,

takes no action and remains in the same state.

3PTY_N01_037 subclause 9.2.4.2, 7th paragraph valid mandatory
Ensure that the IUT in the 3PTY Idle state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), after sending a FACILITY message for CR1 containing a Facility information element with an End3PTY return result component, on receipt of a FACILITY message for CR1 containing a Facility information element with an End3PTY reject component,

takes no action and remains in the same state.

3PTY_N01_038 subclause 9.2.4.2, 7th paragraph valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), after sending a FACILITY or INFORMATION message for CR2 containing a Facility information element with an End3PTY return error component, on receipt of a FACILITY message for CR2 containing a Facility information element with an End3PTY reject component,

takes no action and remains in the same state.

NOTE 14: The choice of 3PTY Active state and the FACILITY message is arbitrary - test could be carried out in any one of a number of 3PTY states and any one of a number of messages could be used. What is most important is that the IUT has just sent an End3PTY return error component.

3PTY_N01_039 subclause 9.2.4.2, 7th paragraph valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), after sending a FACILITY or INFORMATION message for CR1 containing a Facility information element with an End3PTY return error component, on receipt of a FACILITY message for CR1 containing a Facility information element with an End3PTY reject component,
takes no action and remains in the same state.

NOTE 15: The choice of 3PTY Active state and the FACILITY message is arbitrary - test could be carried out in any one of a number of 3PTY states and any one of a number of messages could be used. What is most important is that the IUT has just sent an End3PTY return error component.

3PTY_N01_040 subclause 9.2.5.1 a) valid mandatory
Ensure that the IUT in the 3PTY Active state, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), to clear the call associated with CR1,
sends a DISCONNECT message for CR1 and enters state N12 for CR1.

3PTY_N01_041 subclause 9.2.5.1 a) valid mandatory
Ensure that the IUT in the 3PTY Await Retrieve state, with CR1 in call state N00 and CR2 in call state N10 (Held), after clearing was effected by the sending of a DISCONNECT message for CR1, on receipt of a RETRIEVE message for CR2,
responds with a RETRIEVE ACKNOWLEDGE message for CR2 and enters state N10 (Idle) for CR2 and the 3PTY Idle state.

3PTY_N01_042 subclause 9.2.5.1 b) valid mandatory
Ensure that the IUT, with CR1 in call state N10 (Idle) and CR2 in call state N10 (Held), to clear the call associated with CR2,
sends a DISCONNECT message for CR2 and enters state N12 for CR2.

6.2.2 Remote user procedures

3PTY_N02_001 subclause 9.2.1.1 d) valid mandatory
Ensure that the IUT, to notify a remote user of the successful establishment of a three-way conference,
sends a NOTIFY message to the remote user with a Notification indicator information element coded as "Conference established".

3PTY_N02_002 subclause 9.2.2.1 a) valid mandatory
Ensure that the IUT, to notify a remote user, which was previously in call state N10 (Held), that the other remote user, which was previously in call state N10 (Idle), is disconnected,
sends a NOTIFY message containing a Notification Indicator information element coded as "Remote hold".

3PTY_N02_003 subclause 9.2.2.1 a) valid mandatory
Ensure that the IUT, to notify a remote user that the conference is disconnected,
sends a NOTIFY message with a Notification Indicator information element coded as "Conference disconnected".

7 Compliance

An ATS which complies with this TSS&TP specification shall:

a) consist of a set of test cases corresponding to the set or to a subset of the TPs specified in clause 6;
b) use a TSS which is an appropriate subset of the whole of the TSS specified in clause 5;
c) use the same naming conventions for the test groups and test cases;
d) maintain the relationship specified in clause 6 between the test groups and TPs and the entries in the PICS proforma to be used for test case deselection;
e) comply with ISO/IEC 9646-2 [4].

In the case of a) or b) above, a subset shall be used only where a particular Abstract Test Method (ATM) makes some TPs untestable. All testable TPs from clause 6 shall be included in a compliant ATS.

8 Requirements for a comprehensive testing service

As a minimum the Remote test method, as specified in ISO/IEC 9646-2 [4], shall be used by any organization claiming to provide a comprehensive testing service for network equipment claiming conformance to ETS 300 188-1 [1].

History

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